

Given: $A = \begin{bmatrix} -3 & -9 & 9 & 5 \\ 2 & 6 & -6 & -4 \\ -3 & -9 & 9 & 9 \end{bmatrix}$

REF Form (and RREF): $\text{REF}(A) = \begin{bmatrix} 1 & 3 & -3 & 6 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

a) Basis of the row space:

A basis for the row space is given by the nonzero rows of the REF:

$$\{[1 \ 3 \ -3 \ 0], [0 \ 0 \ 0 \ 1]\}$$

b.) Rank of the matrix:

The rank of a matrix is also the number of nonzero rows of the REF:

Number of nonzero rows is 2, so:

$$\text{rank}(A) = 2$$